

Sentiment Classification Using BERT

情緒分類使用 BERT 神經網路

負面:0 正面:1

負面:0 正面:1 中立:2

負面:0 正面:1 中立:2 無情緒:3

Load model and tokenizer

```
from transformers import AutoTokenizer,
pipeline, BertForSequenceClassification
import torch

# Setting device on GPU if available, else CPU
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print('Using device:', device)
```

Using device: cpu

```
!pip install transformers
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.11/dist-packages (4.51.3)
Requirement already satisfied: filelock in
/usr/local/lib/python3.11/dist-packages (from transformers) (3.18.0)
Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in
/usr/local/lib/python3.11/dist-packages (from transformers) (0.31.2)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.11/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.11/dist-packages (from transformers) (24.2)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.11/dist-packages (from transformers) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.11/dist-packages (from transformers)
(2024.11.6)
Requirement already satisfied: requests in
/usr/local/lib/python3.11/dist-packages (from transformers) (2.32.3)
Requirement already satisfied: tokenizers<0.22,>=0.21 in
/usr/local/lib/python3.11/dist-packages (from transformers) (0.21.1)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.11/dist-packages (from transformers) (0.5.3)
Requirement already satisfied: tqdm>=4.27 in
```

```
/usr/local/lib/python3.11/dist-packages (from transformers) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.11/dist-packages (from huggingface-
hub<1.0,>=0.30.0->transformers) (2025.3.2)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/usr/local/lib/python3.11/dist-packages (from huggingface-
hub<1.0,>=0.30.0->transformers) (4.13.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.11/dist-packages (from requests->transformers)
(3.4.2)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.11/dist-packages (from requests->transformers)
(3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.11/dist-packages (from requests->transformers)
(2.4.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.11/dist-packages (from requests->transformers)
(2025.4.26)
```

```
# You can download the best trained model from huggingface
# https://huggingface.co/clhuang
```

```
# (1) Load model from huggingface
```

```
# model =
```

```
AutoModelForSequenceClassification.from_pretrained("clhuang/albert-
sentiment")
```

```
model = BertForSequenceClassification.from_pretrained("clhuang/albert-
sentiment", num_labels=2) # specify number of labels
```

```
# (2) or Load model from local
```

```
# best_model = "best-model-v1" #
```

```
# # model = AutoModelForSequenceClassification.from_pretrained("./my-
best-model").to(device)
```

```
# model = BertForSequenceClassification.from_pretrained(best_model,
num_labels=2).to(device) # specify number of labels
```

```
model
```

```
BertForSequenceClassification(
```

```
  (bert): BertModel(
```

```
    (embeddings): BertEmbeddings(
```

```
      (word_embeddings): Embedding(21128, 312, padding_idx=0)
```

```
      (position_embeddings): Embedding(512, 312)
```

```
      (token_type_embeddings): Embedding(2, 312)
```

```
      (LayerNorm): LayerNorm((312,), eps=1e-12,
```

```
      elementwise_affine=True)
```

```
      (dropout): Dropout(p=0.0, inplace=False)
```

```
    )
```

```
    (encoder): BertEncoder(
```

```

        (layer): ModuleList(
          (0-3): 4 x BertLayer(
            (attention): BertAttention(
              (self): BertSdpaSelfAttention(
                (query): Linear(in_features=312, out_features=312,
bias=True)
                (key): Linear(in_features=312, out_features=312,
bias=True)
                (value): Linear(in_features=312, out_features=312,
bias=True)
                (dropout): Dropout(p=0.0, inplace=False)
              )
              (output): BertSelfOutput(
                (dense): Linear(in_features=312, out_features=312,
bias=True)
                (LayerNorm): LayerNorm((312,), eps=1e-12,
elementwise_affine=True)
                (dropout): Dropout(p=0.0, inplace=False)
              )
            )
            (intermediate): BertIntermediate(
              (dense): Linear(in_features=312, out_features=1248,
bias=True)
              (intermediate_act_fn): GELUActivation()
            )
            (output): BertOutput(
              (dense): Linear(in_features=1248, out_features=312,
bias=True)
              (LayerNorm): LayerNorm((312,), eps=1e-12,
elementwise_affine=True)
              (dropout): Dropout(p=0.0, inplace=False)
            )
          )
        )
      )
      (pooler): BertPooler(
        (dense): Linear(in_features=312, out_features=312, bias=True)
        (activation): Tanh()
      )
    )
    (dropout): Dropout(p=0.0, inplace=False)
    (classifier): Linear(in_features=312, out_features=2, bias=True)
  )

```

Load model directly

```

from transformers import AutoTokenizer,
AutoModelForSequenceClassification

```

```

tokenizer = AutoTokenizer.from_pretrained("c\u00fclhuang/albert-sentiment")
model =

```

```
AutoModelForSequenceClassification.from_pretrained("clhuang/albert-sentiment")
```

```
# tokenizer
```

```
# tokenizer = AutoTokenizer.from_pretrained(best_model) # or from local
```

```
tokenizer = AutoTokenizer.from_pretrained("clhuang/albert-sentiment")
```

```
#from huggingface
```

```
# tokenizer = BertTokenizer.from_pretrained("bert-base-chinese") # or from huggingface bert-base-chinese
```

```
len(tokenizer)
```

```
21128
```

```
# tokenize can encode text to input_ids and decode input_ids to text  
tokenizer.get_vocab()
```

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'堀': 1827,
'##net': 8914,
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...}
```

```
text="我喜歡"
```

```
# prepare our text into tokenized sequence
```

```
inputs = tokenizer(text)
```

```
inputs
```

```
{'input_ids': [101, 2769, 1599, 3631, 102], 'token_type_ids': [0, 0,
0, 0, 0], 'attention_mask': [1, 1, 1, 1, 1]}
```

```
tokenizer.decode(inputs['input_ids'])
```

```
{"type": "string"}
```

Predict or generate result using pipeline

可能的輸出結果如下：

```
[{'label': 'LABEL_1', 'score': 0.9885562062263489}]  
[{'label': 'LABEL_0', 'score': 0.9052111506462097}]
```

因此需要用到 if 去判斷 label 的值，才能決定 score 是正面還是負面。

若為:LABEL_1 就是正面的 score

若為:LABEL_0 就是負面的 score

```
sentiment_classify = pipeline('sentiment-analysis', model=model,  
tokenizer=tokenizer)
```

Device set to use cpu

```
new_text = '速度很快，昨天下單，今天上午就到啦，看著挺不錯。'  
sentiment_classify(new_text)
```

```
[{'label': 'LABEL_1', 'score': 0.9885562062263489}]
```

```
outputs = sentiment_classify(new_text)  
outputs[0]['score']
```

```
0.9885562062263489
```

```
type(outputs[0]['score'])
```

```
float
```

```
# Positive probability
```

```
round(outputs[0]['score'],2)
```

```
0.99
```

```
# Negative probability
```

```
round(1 - round(outputs[0]['score'],2),2)
```

```
0.01
```

```
new_text = '不喜歡這款產品'  
sentiment_classify(new_text)
```

```
[{'label': 'LABEL_0', 'score': 0.9052111506462097}]
```

Define prediction function using pipeline

```
def get_sentiment_proba(text):
    max_length = 300 # 最多字數 若超出模型訓練時的字數，以模型最大字數為依據
    #max_length = 512 # 最多字數 若超出模型訓練時的字數，以模型最大字數為依據
    outputs = sentiment_classify(text, padding=True,
    max_length=max_length, truncation=True)
    if outputs[0]['label']=='LABEL_1':
        # Get the positive score
        prob_positive = round(outputs[0]['score'],2)
        prob_negative = round(1 - prob_positive, 2)
    else:
        # Calculate the negative score
        prob_negative = round(outputs[0]['score'],2)
        prob_positive = round(1 - prob_negative, 2)

    response = {'Negative':prob_negative, 'Positive': prob_positive}
    return response

new_text = '速度很快，昨天下單，今天上午就到啦，看著挺不錯。'
get_sentiment_proba( new_text )

{'Negative': 0.01, 'Positive': 0.99}

new_text = '已經買了這種蘋果好多次了，寶寶喜歡上了這款蘋果，一直選擇這款'
get_sentiment_proba( new_text )

{'Negative': 0.02, 'Positive': 0.98}

new_text = '不喜歡這款產品'

get_sentiment_proba( new_text )

{'Negative': 0.91, 'Positive': 0.09}

new_text = '非常不喜歡這款產品'

get_sentiment_proba( new_text )

{'Negative': 0.72, 'Positive': 0.28}
```

Define prediction function using model or model.generate()

```
## Prediction
target_names=['Negative','Positive']
max_length = 200 # 最多字數 若超出模型訓練時的字數，以模型最大字數為依據
```

```

def get_sentiment_proba_from_model(text):
    # prepare our text into tokenized sequence
    inputs = tokenizer(text, padding=True, truncation=True,
max_length=max_length, return_tensors="pt").to(device)
    # perform inference to our model
    outputs = model(**inputs)
    # get output probabilities by doing softmax
    probs = outputs[0].softmax(1)

    response = {'Negative': round(float(probs[0, 0]), 2), 'Positive':
round(float(probs[0, 1]), 2)}
    # executing argmax function to get the candidate label
    #return probs.argmax()
    return response

new_text = '不喜歡這款產品'

get_sentiment_proba_from_model( new_text )

{'Negative': 0.91, 'Positive': 0.09}

```

Pediction 模型使用

label <--> id

```

# Map labels to integers
categories=['負面', '正面']

label_to_id = { cate : i for i, cate in enumerate(categories)}
label_to_id
{'負面': 0, '正面': 1}

id_to_label = { i : cate for i, cate in enumerate(categories)}
id_to_label
{0: '負面', 1: '正面'}

# Function to make predictions
def predict_sentiment(text, model, tokenizer, device):
    max_length = 512 # 最多字數 若超出模型訓練時的字數，以模型最大字數為依據
    # Tokenize the input text
    inputs = tokenizer(
        text,

```

```

        max_length=max_length,
        truncation=True,
        return_tensors="pt"
    ).to(device)

    # Get model predictions
    with torch.no_grad():
        outputs = model(**inputs)

    # Extract logits and apply softmax to get probabilities
    # logits = outputs.logits
    logits = outputs["logits"] # 取出logits

    probabilities = torch.nn.functional.softmax(logits, dim=-1)

    # Get the predicted class (0: negative, 1: positive)
    predicted_class = torch.argmax(probabilities, dim=-1).item()

    # Get the class name using id_to_label
    predicted_label = id_to_label[predicted_class]

    # Get the confidence score
    confidence = probabilities[0][predicted_class].item()

    return {
        "text": text,
        "sentiment": predicted_label,
        "confidence": round(confidence,2),
        "probabilities": {
            id_to_label[i]: round(prob.item(),2) for i, prob in
enumerate(probabilities[0])
        }
    }

text = "今天天氣真好，我很開心"
predict_sentiment(text, model, tokenizer, device)

{'text': '今天天氣真好，我很開心',
 'sentiment': '正面',
 'confidence': 0.81,
 'probabilities': {'負面': 0.19, '正面': 0.81}}

text = "這個產品品質差，服務更糟糕"
predict_sentiment(text, model, tokenizer, device)

{'text': '這個產品品質差，服務更糟糕',
 'sentiment': '負面',
 'confidence': 0.99,
 'probabilities': {'負面': 0.99, '正面': 0.01}}

```

```
text = "這家餐廳的食物美味，環境也很舒適"  
predict_sentiment(text, model, tokenizer, device)  
  
{  
  'text': '這家餐廳的食物美味，環境也很舒適',  
  'sentiment': '正面',  
  'confidence': 0.98,  
  'probabilities': {'負面': 0.02, '正面': 0.98}}  
}
```